

ROLL NO.

2501921330291

G.L. BAJAJ INSTITUTE OF TECHNOLOGY & MANAGEMENT
GREATER NOIDA

B.Tech (CSE, CSDS, CS-AIML, CSAI, CSIT) 2nd Year

Sessional Test- 1 (ODD SEM 2026-27)

Subject Name: Data Structure

Subject Code: BCS301

Duration: 2:00 Hrs

Max. Marks: 50

- Note:
- (i) No student will be allowed to leave the examination Room before end of the exam.
 - (ii) Diagrams should be neat and clean.
 - (iii) Mention question number/section correctly.
 - (iv) Be precise in your answer.
 - (v) Do not write anything on question paper except Roll number.

Course Outcomes:

Following are the course outcomes of the subject :-

CO Code	Course Outcome (CO)	Bloom's Level
BCS301.1	Describe how arrays, linked lists, stacks, queues, trees, and graphs are represented in memory, used by the algorithms and their common applications.	K1(Remember), K2(Understand)
BCS301.2	Discuss the computational efficiency of the sorting and searching algorithms.	K2(Understand)
BCS301.3	Implementation of Trees and Graphs and perform various operations on these data structure.	Apply(K3)
BCS301.4	Understanding the concept of recursion, application of recursion and its implementation and removal of recursion.	Analyze(K4)
BCS301.5	Identify the alternative implementations of data structures with respect to its performance to solve a real-world problem.	Evaluate(K5), Create(K6)

Section: A

(5*2 = 10)				
1. Attempt all questions.		Marks	CO	BL
Q. No.	Questions			
a)	Describe indexed sequential search.	2	BCS301.2	K2
b)	Make a comparison between linked list and linear array.	2	BCS301.1	K2
c)	Explain the concept of non linear data structure with the help of example.	2	BCS301.1	K2
d)	Prove that $3n^2 + 2n = O(n^2)$.	2	BCS301.2	K2
e)	What does the following recursive function do for a given Linked List with first node as head? void fun1(struct node* head) { if(head == NULL) return; fun1(head->next); printf("%d ", head->data); } 	2	BCS301.1	K2

Section: B

(4*5 = 20)				
2. Attempt any four of the following:		Marks	CO	BL
Q. No.	Questions			
a)	What is a Stack? Write a C program to reverse a string using stack.	5	BCS301.1	K3
b)	An array ARR[-7...15, 15...25] stores elements in Row Major Wise with each element requiring 4 bytes of storage. Find the address of ARR[10][15] when the base address is 3500.	5	BCS301.1	K3

e)	Describe Sparse matrix. Explain how the sparse matrix is stored efficiently in memory with example.	5	BCS301.1	K2
d)	Define Circular queue. Write a C functions to insert and delete an element in circular queue.	5	BCS301.1	K2
e)	Explain various types of asymptotic notations.	5	BCS301.1	K3
f)	i) Find the result of the following postfix expression using stack 2 3 9 * + 2 3 ^ - 6 2 / + , show the contents of each and every steps. Where ^ is an exponent operator. ii) Find the result of given prefix expression: + - 7 2 * 8 / 12 4	5	BCS301.1	K3

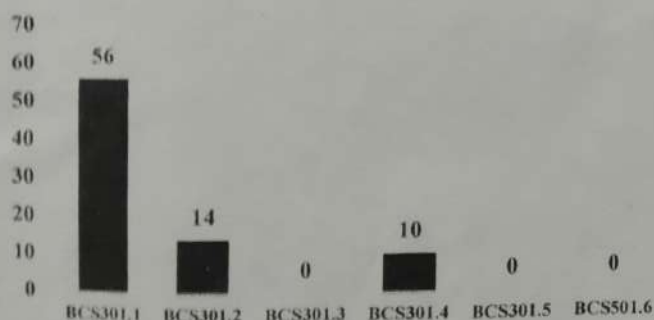
Section: C

3. Attempt any one question: (1*10 = 10)				
Q. No.	Questions	Marks	CO	BL
a)	Write an algorithm for converting infix expression into postfix expression using stack. Apply your algorithm to convert the given infix expression into its equivalent postfix expression. $A+(B*C-(D/E^F)*G)*H$	10	BCS301.1	K3
b)	Discuss doubly linked list. Write a C function for insertion and deletion of a node from starting and ending position in a doubly linked list.	10	BCS301.1	K2

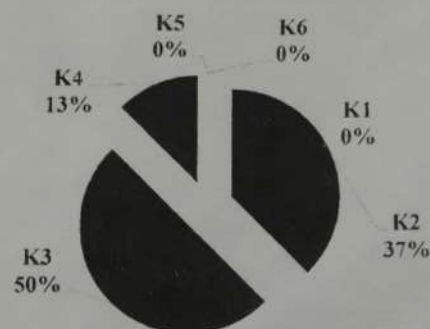
Section: D

4. Attempt any one question: (1*10 = 10)				
Q. No.	Questions	Marks	CO	BL
a)	Construct a Hash table that contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key \% 10}$. Insert values 43, 165, 62, 123, 142 in the table. Show all the steps and find the location of the key value 142?	10	BCS301.2	K3
b)	Write a recursive function to implement binary search and analyze the time complexity of the algorithm.	10	BCS301.4	K4

Course Outcome Wise Marks Distribution



Bloom's Level Distribution



Checked By
(Head of the Department)